Part 1: Synthesis & Decomposition Reactions

1. Hydrogen gas reacts with oxygen gas to form water.

a) $H_2\ (g) + O_2\ (g) \rightarrow H_2O\ (l)$ (Unbalanced)

b) $2\ H_2\ (g) + O_2\ (g) \rightarrow 2\ H_2O\ (l)$ (**Correct**)

c) $H_2\ (g) + 2\ O_2\ (g) \rightarrow H_2O_2\ (l)$ (Incorrect product)

d) $4\ H_2\ (g) + 2\ O_2\ (g) \rightarrow 3\ H_2O\ (l)$ (Incorrect coefficients)

2. Solid magnesium reacts with oxygen gas to form magnesium oxide.

a) $Mg\ (s) + O_2\ (g) \rightarrow MgO\ (s)$ (Unbalanced)

b) $2\ Mg\ (s) + 2\ O_2\ (g) \rightarrow 2\ MgO\ (s)$ (Unbalanced oxygen)

c) $2\ Mg\ (s) + O_2\ (g) \rightarrow 2\ MgO\ (s)$ (**Correct**)

d) $Mg\ (s) + 2\ O_2\ (g) \rightarrow MgO_2\ (s)$ (Incorrect product formula)

3. Solid aluminum reacts with chlorine gas to form aluminum chloride.

a) $2\ Al\ (s) + 3\ Cl_2\ (g) \rightarrow 2\ AlCl_3\ (s)$ (**Correct**)

b) $Al\ (s) + Cl_2\ (g) \rightarrow AlCl_3\ (s)$ *(Unbalanced)*

c) $Al\ (s) + Cl_2\ (g) \rightarrow AlCl_2\ (s)$ (Incorrect formula for aluminum chloride)

d) $4\ Al\ (s) + 3\ Cl_2\ (g) \rightarrow 4\ AlCl_3\ (s)$ (Unbalanced coefficients)

4. Solid iron reacts with oxygen gas to form iron(III) oxide (rust).

a) $Fe\ (s) + O_2\ (g) \rightarrow FeO\ (s)$ (Incorrect iron oxidation state)

b) $2\ Fe\ (s) + 3\ O_2\ (g) \rightarrow Fe_2O_3\ (s)$ *(Unbalanced iron)*

c) $Fe\ (s) + O_2\ (g) \rightarrow Fe_2O_3\ (s)$ *(Unbalanced)*

d) $4\ Fe\ (s) + 3\ O_2\ (g) \rightarrow 2\ Fe_2O_3\ (s)$ (**Correct**)

5. Solid calcium oxide reacts with water to form solid calcium hydroxide.

a) $CaO\ (s) + H_2O\ (l) \rightarrow Ca(OH)_2\ (s)$ (**Correct** and already balanced)

b) $CaO\ (s) + H_2O\ (l) \rightarrow CaOH\ (s)$ (Incorrect formula)

c) $2\ CaO\ (s) + H_2O\ (l) \rightarrow Ca(OH)_2\ (s)$ (Unbalanced calcium)

d) $CaO\ (s) + 2\ H_2O\ (l) \rightarrow Ca_2OH\ (s)$ (Incorrect formula and coefficients)

6. The decomposition of solid water (ice/steam) into hydrogen and oxygen gases.

a) $2\ H_2O\ (g) \rightarrow 2\ H_2\ (g) + O_2\ (g)$ (**Correct**)

b) $H_2O\ (g) \rightarrow H_2\ (g) + O_2\ (g)$ (Unbalanced oxygen)

c) $2\ H_2O\ (g) \rightarrow H_2\ (g) + 2\ O_2\ (g)$ (Incorrect oxygen count)

d) $H_2O\ (g) \rightarrow H\ (g) + O\ (g)$ (Ignoring diatomic elements)

7. Solid potassium chlorate decomposes into potassium chloride and oxygen gas.

a) $KClO_3\ (s) \rightarrow KCl\ (s) + O_2\ (g)$ (Unbalanced oxygen)

b) $2\ KClO_3\ (s) \rightarrow 2\ KCl\ (s) + 3\ O_2\ (g)$ ***(Correct)***

c) $KClO_3\ (s) \rightarrow K\ (s) + Cl_2\ (g) + O_2\ (g)$ (Incorrect decomposition products)

d) $3\ KClO_3\ (s) \rightarrow 3\ KCl\ (s) + O_2\ (g)$ (Unbalanced oxygen)

Part 2: Single Replacement & Metal Reactions

8. Zinc metal reacts with *aq*ueous hydrochloric acid to produce hydrogen gas and *aq*ueous zinc chloride.

a) $Zn\ (s) + HCl\ (aq) \rightarrow H_2\ (g) + ZnCl\ (aq)$ *(Incorrect zinc chloride formula)*

b) $Zn\ (s) + 2\ HCl\ (aq) \rightarrow H_2\ (g) + ZnCl_2\ (aq)$ ***(Correct)***

c) $2\ Zn\ (s) + HCl\ (aq) \rightarrow H_2\ (g) + 2\ ZnCl_2\ (aq)$ (Unbalanced)

d) $Zn\ (s) + 2\ HCl\ (aq) \rightarrow Cl_2\ (g) + ZnH_2\ (aq)$ (Incorrect products)

9. Solid aluminum reacts with *aq*ueous copper(II) sulfate to produce solid copper and *aq*ueous aluminum sulfate.

a) $Al\ (s) + CuSO_4\ (aq) \rightarrow Cu\ (s) + Al_2(SO_4)_3\ (aq)$ (Unbalanced)

b) $Al\ (s) + CuSO_4\ (aq) \rightarrow Cu\ (s) + AlSO_4\ (aq)$ (*Incorrect aluminum sulfate formula*)

c) $2\ Al\ (s) + 3\ CuSO_4\ (aq) \rightarrow 3\ Cu\ (s) + Al_2(SO_4)_3\ (aq)$ (***Correct***)

d) $3\ Al\ (s) + 2\ CuSO_4\ (aq) \rightarrow 2\ Cu\ (s) + Al_3(SO_4)_2\ (aq)$ (*Incorrect formula and ratios*)

10. Solid iron reacts with *aq*ueous lead(II) nitrate to produce solid lead and *aq*ueous iron(II) nitrate.

a) $Fe\ (s) + Pb(NO_3)_2\ (aq) \rightarrow Pb\ (s) + Fe(NO_3)_2\ (aq)$ (***Correct** and balanced*)

b) $2\ Fe\ (s) + Pb(NO_3)_2\ (aq) \rightarrow Pb\ (s) + 2\ Fe(NO_3)_2\ (aq)$ (*Unbalanced iron*)

c) $Fe\ (s) + 2\ Pb(NO_3)_2\ (aq) \rightarrow 2\ Pb\ (s) + Fe(NO_3)_4\ (aq)$ (*Incorrect iron oxidation state*)

d) $Fe\ (s) + Pb(NO_3)_2\ (aq) \rightarrow Pb\ (s) + FeNO_3\ (aq)$ (*Incorrect formula*)

11. Solid magnesium reacts with *aq*ueous silver nitrate to produce solid silver and *aq*ueous magnesium nitrate.

a) $Mg\ (s) + AgNO_3\ (aq) \rightarrow Ag\ (s) + Mg(NO_3)_2\ (aq)$ (*Unbalanced*)

b) $2\ Mg\ (s) + AgNO_3\ (aq) \rightarrow Ag\ (s) + 2\ Mg(NO_3)_2\ (aq)$ (*Unbalanced*)

c) $Mg\ (s) + AgNO_3\ (aq) \rightarrow Ag\ (s) + MgNO_3\ (aq)$ (*Incorrect formula*)

d) $Mg\ (s) + 2\ AgNO_3\ (aq) \rightarrow 2\ Ag\ (s) + Mg(NO_3)_2\ (aq)$ (***Correct***)

12. Solid lithium reacts with liquid water to produce hydrogen gas and *aq*ueous lithium hydroxide.

a) $Li\ (s) + H_2O\ (l) \rightarrow H_2\ (g) + LiOH\ (aq)$ (*Unbalanced*)

b) $2\ Li\ (s) + 2\ H_2O\ (l) \rightarrow H_2\ (g) + 2\ LiOH\ (aq)$ (***Correct***)

c) $Li\ (s) + 2\ H_2O\ (l) \rightarrow H_2\ (g) + Li_2OH\ (aq)$ (*Incorrect formula*)

d) $4\ Li\ (s) + 2\ H_2O\ (l) \rightarrow 2\ H_2\ (g) + 4\ LiOH\ (aq)$ (*Unbalanced coefficients*)

Part 3: Double Replacement & Precipitation Reactions

13. *Aq*ueous barium chloride reacts with *aq*ueous sodium sulfate to form a precipitate of barium sulfate and *aq*ueous sodium chloride.

a) $BaCl_2\ (aq) + Na_2SO_4\ (aq) \rightarrow BaSO_4\ (s) + NaCl\ (aq)$ (*Unbalanced sodium/chlorine*)

b) $2\ BaCl_2\ (aq) + Na_2SO_4\ (aq) \rightarrow 2\ BaSO_4\ (s) + NaCl\ (aq)$ (*Unbalanced barium*)

c) $BaCl_2\ (aq) + Na_2SO_4\ (aq) \rightarrow Ba_2SO_4\ (s) + NaCl_2\ (aq)$ (*Incorrect formulas*)

d) $BaCl_2\ (aq) + Na_2SO_4\ (aq) \rightarrow BaSO_4\ (s) + 2\ NaCl\ (aq)$ (***Correct***)

14. *Aq*ueous silver nitrate reacts with *aq*ueous sodium chloride to form solid silver chloride and *aq*ueous sodium nitrate.

a) $2\ AgNO_3\ (aq) + NaCl\ (aq) \rightarrow 2\ AgCl\ (s) + Na_2NO_3\ (aq)$ (*Unbalanced and incorrect formula*)

b) $AgNO_3\ (aq) + 2\ NaCl\ (aq) \rightarrow AgCl_2\ (s) + Na_2NO_3\ (aq)$ (*Incorrect formulas*)

c) $AgNO_3\ (aq) + NaCl\ (aq) \rightarrow AgCl\ (s) + NaNO_3\ (aq)$ (***Correct** and balanced*)

d) $2\ AgNO_3\ (aq) + 2\ NaCl\ (aq) \rightarrow 2\ AgCl\ (s) + Na_2NO_3\ (aq)$ (*Unbalanced sodium/oxygen*)

15. *Aq*ueous lead(II) nitrate reacts with *aq*ueous potassium iodide to form solid lead(II) iodide and *aq*ueous potassium nitrate.

a) $Pb(NO_3)_2\ (aq) + KI\ (aq) \rightarrow PbI_2\ (s) + KNO_3\ (aq)$ (*Unbalanced*)

b) $Pb(NO_3)_2\ (aq) + 2\ KI\ (aq) \rightarrow PbI_2\ (s) + 2\ KNO_3\ (aq)$ (***Correct***)

c) $2\ Pb(NO_3)_2\ (aq) + KI\ (aq) \rightarrow PbI_2\ (s) + 2\ KNO_3\ (aq)$ (*Unbalanced lead*)

d) $Pb(NO_3)_2\ (aq) + KI\ (aq) \rightarrow PbI + K_2NO_3\ (aq)$ (*Incorrect formulas*)

16. *Aq*ueous copper(II) sulfate reacts with *aq*ueous sodium hydroxide to form solid copper(II) hydroxide and *aq*ueous sodium sulfate.

a) $CuSO_4\ (aq) + NaOH\ (aq) \rightarrow Cu(OH)_2\ (s) + Na_2SO_4\ (aq)$ (*Unbalanced*)

b) $CuSO_4\ (aq) + 2\ NaOH\ (aq) \rightarrow Cu(OH)_2\ (s) + Na_2SO_4\ (aq)$ (***Correct***)

c) $2\ CuSO_4\ (aq) + NaOH\ (aq) \rightarrow Cu(OH)_2\ (s) + 2\ Na_2SO_4\ (aq)$ (*Unbalanced copper*)

d) $CuSO_4\ (aq) + NaOH\ (aq) \rightarrow CuOH\ (s) + NaSO_4\ (aq)$ (*Incorrect formulas*)

Part 4: Neutralization Reactions

17. *Aq*ueous hydrochloric acid reacts with *aq*ueous sodium hydroxide to form liquid water and *aq*ueous sodium chloride.

a) $2\ HCl\ (aq) + NaOH\ (aq) \rightarrow H_2O\ (l) + Na_2Cl_2\ (aq)$ (*Incorrect formula*)

b) $HCl\ (aq) + 2\ NaOH\ (aq) \rightarrow 2\ H_2O\ (l) + NaCl_2\ (aq)$ (*Incorrect formula*)

c) $HCl\ (aq) + NaOH\ (aq) \rightarrow H_2\ (g) + NaClO\ (aq)$ (*Incorrect products*)

d) $HCl\ (aq) + NaOH\ (aq) \rightarrow H_2O\ (l) + NaCl\ (aq)$ (***Correct** and balanced*)

18. *Aq*ueous sulfuric acid reacts with *aq*ueous sodium hydroxide to form liquid water and *aq*ueous sodium sulfate.

a) $H_2SO_4\ (aq) + NaOH\ (aq) \rightarrow H_2O\ (l) + Na_2SO_4\ (aq)$ (*Unbalanced*)

c) $2\ H_2SO_4\ (aq) + NaOH\ (aq) \rightarrow H_2O\ (l) + Na_2SO_4\ (aq)$ (*Unbalanced hydrogen*)

b) $H_2SO_4\ (aq) + 2NaOH\ (aq) \rightarrow 2\ H_2O\ (l) + Na_2SO_4\ (aq)$ (***Correct***)

d) $H_2SO_4\ (aq) + NaOH\ (aq) \rightarrow H_2O\ (l) + NaSO_4\ (aq)$ (*Incorrect formula*)

19. *Aq*ueous phosphoric acid reacts with *aq*ueous potassium hydroxide to form liquid water and *aq*ueous potassium phosphate.

a) $H_3PO_4\ (aq) + KOH\ (aq) \rightarrow H_2O\ (l) + K_3PO_4\ (aq)$ (*Unbalanced*)

b) $H_3PO_4\ (aq) + 3\ KOH\ (aq) \rightarrow 3\ H_2O\ (l) + K_3PO_4\ (aq)$ (***Correct***)

c) $3\ H_3PO_4\ (aq) + KOH\ (aq) \rightarrow H_2O\ (l) + K_3PO_4\ (aq)$ (*Unbalanced hydrogen/phosphate*)

d) $H_3PO_4\ (aq) + 3\ KOH\ (aq) \rightarrow H_2O\ (l) + K_3PO_4\ (aq)$ (*Unbalanced water count*)

20. *Aq*ueous nitric acid reacts with *aq*ueous calcium hydroxide to form liquid water and *aq*ueous calcium nitrate.

a) $HNO_3\ (aq) + Ca(OH)_2\ (aq) \rightarrow H_2O\ (l) + Ca(NO_3)_2\ (aq)$ (*Unbalanced*)

b) $HNO_3\ (aq) + 2\ Ca(OH)_2\ (aq) \rightarrow 2\ H_2O\ (l) + Ca(NO_3)_2\ (aq)$ (*Unbalanced calcium*)

c) $2\ HNO_3\ (aq) + Ca(OH)_2\ (aq) \rightarrow H_2O\ (l) + Ca(NO_3)_2\ (aq)$ (*Unbalanced hydrogen/oxygen*)

d) $2\ HNO_3\ (aq) + Ca(OH)_2\ (aq) \rightarrow 2\ H_2O\ (l) + Ca(NO_3)_2\ (aq)$ (***Correct***)

Part 5: Combustion Reactions

21. Combustion of methane gas (CH_4) **in excess oxygen gas.**

a) $CH_4\ (g) + O_2\ (g) \rightarrow CO_2\ (g) + H_2O\ (g)$ (*Unbalanced*)

b) $CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2\ (g) + 2\ H_2O\ (g)$ (***Correct***)

c) $CH_4\ (g) + O_2\ (g) \rightarrow CO\ (g) + 2\ H_2O\ (g)$ (*Incomplete combustion product*)

d) $2\ CH_4\ (g) + O_2\ (g) \rightarrow 2\ CO_2\ (g) + 4\ H_2O\ (g)$ (*Unbalanced oxygen*)

22. Combustion of propane gas (C_3H_8) **in excess oxygen gas.**

a) $C_3H_8\ (g) + O_2\ (g) \rightarrow CO_2\ (g) + H_2O\ (g)$ (*Unbalanced*)

b) $C_3H_8\ (g) + 5\ O_2\ (g) \rightarrow 3\ CO_2\ (g) + 4\ H_2O\ (g)$ (***Correct***)

c) $C_3H_8\ (g) + 3\ O_2\ (g) \rightarrow 3\ CO_2\ (g) + 4\ H_2O\ (g)$ (*Unbalanced oxygen*)

d) $2\ C_3H_8\ (g) + 7\ O_2\ (g) \rightarrow 6\ CO_2\ (g) + 8\ H_2O\ (g)$ (*Unbalanced coefficients*)

23. Combustion of ethanol liquid (C_2H_5OH) **in excess oxygen gas.**

a) $C_2H_5OH\ (l) + 3\ O_2\ (g) \rightarrow 2\ CO_2\ (g) + 3\ H_2O\ (g)$ (***Correct***)

b) $C_2H_5OH\ (l) + O_2\ (g) \rightarrow CO_2\ (g) + H_2O\ (g)$ (*Unbalanced*)

c) $C_2H_5OH\ (l) + 2\ O_2\ (g) \rightarrow 2\ CO_2\ (g) + 2\ H_2O\ (g)$ (*Unbalanced oxygen/hydrogen*)

d) $2\ C_2H_5OH\ (l) + 3\ O_2\ (g) \rightarrow 4\ CO_2\ (g) + 6\ H_2O\ (g)$ (*Unbalanced oxygen*)

24. Combustion of liquid benzene (C_6H_6) **in oxygen gas.**

a) $C_6H_6\ (l) + O_2\ (g) \rightarrow 6\ CO_2\ (g) + 3\ H_2O\ (g)$ (*Unbalanced*)

b) $C_6H_6\ (l) + 15\ O_2\ (g) \rightarrow 6\ CO_2\ (g) + 6\ H_2O\ (g)$ (*Unbalanced carbon*)

c) $2\ C_6H_6\ (l) + O_2\ (g) \rightarrow 12\ CO_2\ (g) + 6\ H_2O\ (g)$ (*Unbalanced oxygen*)

d) $2\ C_6H_6\ (l) + 15\ O_2\ (g) \rightarrow 12\ CO_2\ (g) + 6\ H_2O\ (g)$ (***Correct***)

25. Combustion of solid glucose $(C_6H_{12}O_6)$ **in oxygen gas (cellular respiration).**

a) $C_6H_{12}O_6\ (s) + O_2\ (g) \rightarrow 6\ CO_2\ (g) + 6\ H_2O\ (g)$ (*Unbalanced*)

b) $C_6H_{12}O_6\ (s) + 6\ O_2\ (g) \rightarrow 6\ CO_2\ (g) + 6\ H_2O\ (g)$ (***Correct***)

c) $6\ C_6H_{12}O_6\ (s) + O_2\ (g) \rightarrow CO_2\ (g) + H_2O\ (g)$ (*Unbalanced coefficients*)

d) $C_6H_{12}O_6\ (s) + 12\ O_2\ (g) \rightarrow 6\ CO_2\ (g) + 12\ H_2O\ (g)$ (*Unbalanced oxygen*)